

Problem

An op amp differentiator has $R = 250 \text{ k}\Omega$ and $C = 10 \text{ }\mu\text{F}$. The input voltage is a ramp $r(t) = 7t \text{ mV}$. Find the output voltage.

Step-by-step solution

Step 1 of 2

The input voltage is given as,

$$\begin{aligned} v_i(t) &= r(t) \\ &= 7t \text{ mV} \end{aligned}$$

Draw the circuit for an op amp differentiator.

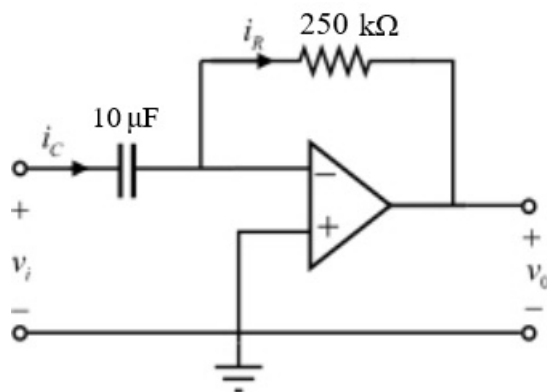


Figure 1

Step 2 of 2

Refer to equation 6.37 in the textbook.

$$v_o = -RC \frac{dv_i}{dt}$$

Substitute $250 \text{ k}\Omega$ for R , $10 \text{ }\mu\text{F}$ for C and $7t \text{ mV}$ for v_i .

$$\begin{aligned} v_o &= -(250 \times 10^3 \times 10 \times 10^{-6}) \frac{d}{dt} (7t \times 10^{-3}) \\ &= -250 \times 10^3 \times 10 \times 10^{-6} \times 7 \times 10^{-3} \\ &= -17.5 \times 10^{-3} \text{ V} \end{aligned}$$

Therefore, the output voltage of the differentiator is -17.5 mV